

Chapter 4 – Energy and Potential

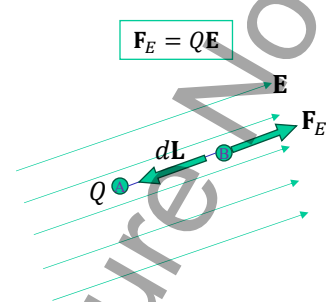
| No. | Course Learning Outcome (CLO) Statements                                                                                                                     | Assessment Tools | Taxonomy Levels | PLO |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|-----|
| 2   | Formulate electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics. | Q2, M1, M2, F    | C5              | 2   |

Objectives:

- 1. Concept of Work Done and Definition of Potential
- 2. Potential Field due to Different Charge Distributions
- 3. Concept of Conservative field and Gradient of that field
- 4. Potential Energy stored in system of charges

Work Done in the presence of E field

Suppose a point charge  $Q$  is to be moved a distance  $d\mathbf{L}$  in the presence of an electric field  $\mathbf{E}$ .



$\mathbf{F}_E = Q\mathbf{E}$

This force arises from the field  $\mathbf{E}$  moving the charge from  $A$  to  $B$ .  
Now, the component of this force along  $d\mathbf{L} = dL \mathbf{a}_L$  can be found as,

$F_{EL} = \mathbf{F}_E \cdot \mathbf{a}_L = Q\mathbf{E} \cdot \mathbf{a}_L$

If we attempt to move the test charge against the  $\mathbf{E}$  field, the applied force must be equal and opposite to the force associated with the field,

$F_{appl} = -F_{EL} = -Q\mathbf{E} \cdot \mathbf{a}_L$

Total Energy Expended is the product of force and distance.

$dW = F_{appl}dL = -Q\mathbf{E} \cdot \mathbf{a}_L dL$

$dW = -Q\mathbf{E} \cdot d\mathbf{L}$

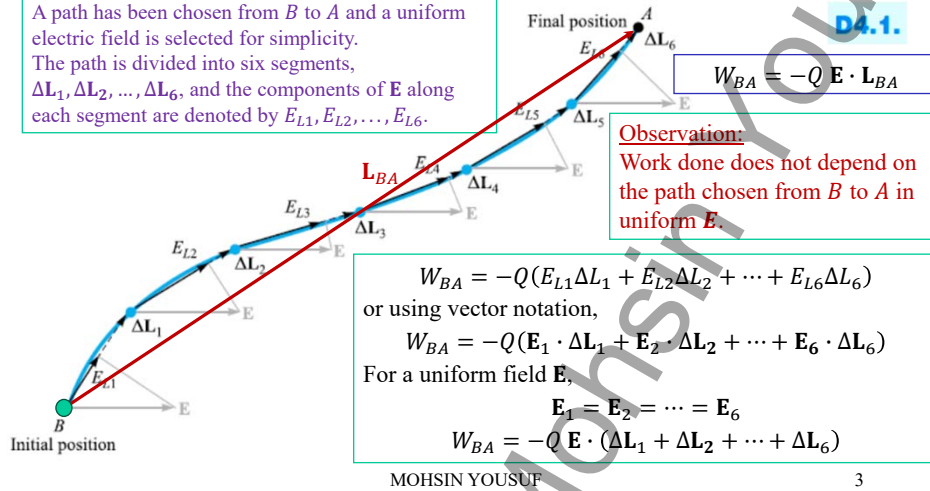
This differential amount of work required may be zero under several conditions:  
1)  $\mathbf{E}$ ,  $Q$  or  $d\mathbf{L}$  is zero.  
2)  $\mathbf{E}$  and  $d\mathbf{L}$  are perpendicular

## Work Done in the presence of $\mathbf{E}$ field (Line Integral)

The work required to move the charge  $Q$  a finite distance can be determined by,

$$W = \int dW = -Q \int_{init}^{final} \mathbf{E} \cdot d\mathbf{L}$$

A path has been chosen from  $B$  to  $A$  and a uniform electric field is selected for simplicity. The path is divided into six segments,  $\Delta\mathbf{L}_1, \Delta\mathbf{L}_2, \dots, \Delta\mathbf{L}_6$ , and the components of  $\mathbf{E}$  along each segment are denoted by  $E_{L1}, E_{L2}, \dots, E_{L6}$ .



## Work Done in the presence of $\mathbf{E}$ field (Line Integral)

Example 4.1: Given a non-uniform field,  
 $\mathbf{E} = y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z$

$$W = \int dW = -Q \int_{init}^{final} \mathbf{E} \cdot d\mathbf{L}$$

Determine the work done in carrying  $2C$  of charge from  $B(1, 0, 1)$  to  $A(0.8, 0.6, 1)$  along the shorter arc of the circle

$$x^2 + y^2 = 1, \quad z = 1$$

In RCS,  
 $d\mathbf{L} = dx\mathbf{a}_x + dy\mathbf{a}_y + dz\mathbf{a}_z$

The integrand  $\mathbf{E} \cdot d\mathbf{L}$  is,  
 $\mathbf{E} \cdot d\mathbf{L} = (y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z) \cdot (dx\mathbf{a}_x + dy\mathbf{a}_y + dz\mathbf{a}_z)$

The expended work can now be computed as,

$$W = -Q \int_B^A \mathbf{E} \cdot d\mathbf{L}$$

$$W = -2 \int_B^A (y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z) \cdot (dx\mathbf{a}_x + dy\mathbf{a}_y + dz\mathbf{a}_z)$$

$$W = -2 \int_{x=1}^{0.8} y dx - 2 \int_{y=0}^{0.6} x dy - 4 \int_{z=1}^1 dz$$

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### Work Done in the presence of $\mathbf{E}$ field (Line Integral)

Example 4.1: Given a non-uniform field,  
 $\mathbf{E} = y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z$

$$W = \int dW = -Q \int_{init}^{final} \mathbf{E} \cdot d\mathbf{L}$$

Determine the work done in carrying  $2C$  of charge from  $B(1, 0, 1)$  to  $A(0.8, 0.6, 1)$  along the shorter arc of the circle

$$x^2 + y^2 = 1, \quad z = 1$$

The expended work can now be computed as,

$$\begin{aligned} W &= -2 \int_{x=1}^{0.8} y dx - 2 \int_{y=0}^{0.6} x dy - 4 \int_{z=1}^1 dz \\ W &= -2 \int_{x=1}^{0.8} \sqrt{1-x^2} dx - 2 \int_{y=0}^{0.6} \sqrt{1-y^2} dy - 0 \\ W &= -\left[x\sqrt{1-x^2} + \sin^{-1} x\right]_1^{0.8} - \left[y\sqrt{1-y^2} + \sin^{-1} y\right]_0^{0.6} \\ W &= -(0.48 + 0.927 - 0 - 1.571) - (0.48 + 0.644 - 0 - 0) \\ W &= -0.96 \text{ J} \end{aligned}$$

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### Work Done in the presence of $\mathbf{E}$ field (Line Integral)

Example 4.2: Given a non-uniform field,  
 $\mathbf{E} = y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z$

$$W = \int dW = -Q \int_{init}^{final} \mathbf{E} \cdot d\mathbf{L}$$

Determine the work done in carrying  $2C$  of charge from  $B(1, 0, 1)$  to  $A(0.8, 0.6, 1)$  along a straight line from  $B$  to  $A$ .

Two point slope formula,

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

Relating  $x$  and  $y$ ,

$$\begin{aligned} y - 0 &= \frac{0.6 - 0}{0.8 - 1} (x - 1) \\ y &= \frac{0.6}{-0.2} (x - 1) = -3(x - 1) \\ \text{and, } x &= 1 - \frac{y}{3} \end{aligned}$$

Relating  $y$  and  $z$ ,

$$\begin{aligned} z - 1 &= \frac{1 - 1}{0.6 - 1} (y - 0) \\ z - 1 &= 0, \quad z = 1 \end{aligned}$$

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### Work Done in the presence of $\mathbf{E}$ field (Line Integral)

Example 4.2: Given a non-uniform field,  
 $\mathbf{E} = y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z$

$$W = \int dW = -Q \int_B^A \mathbf{E} \cdot d\mathbf{L}$$

Determine the work done in carrying 2C of charge from  $B(1, 0, 1)$  to  $A(0.8, 0.6, 1)$  along a straight line from  $B$  to  $A$ .

$$y = -3(x - 1), \quad x = \left(1 - \frac{y}{3}\right), \quad \text{and } z = 1$$

The expended work can now be computed as,

$$\begin{aligned} W &= -2 \int_{x=1}^{0.8} y dx - 2 \int_{y=0}^{0.6} x dy - 4 \int_{z=1}^1 dz \\ W &= 6 \int_{x=1}^{0.8} (x - 1) dx - 2 \int_{y=0}^{0.6} \left(1 - \frac{y}{3}\right) dy \\ W &= -0.96 \text{ J} \end{aligned}$$

**Observation:** This is the same answer we found using the circular path between the same two points. It demonstrates that the work done is independent of the path taken in any electrostatic field. Line integrals are path independent.

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### Work Done in the presence of $\mathbf{E}$ field (Line Integral)

$$W = -Q \int_B^A \mathbf{E} \cdot d\mathbf{L}$$

**D4.2.** Calculate the work done in moving a 4-C charge from  $B(1, 0, 0)$  to  $A(0, 2, 0)$  along the path  $y = 2 - 2x$ ,  $z = 0$  in the field  $\mathbf{E} = (a) 5\mathbf{a}_x \text{ V/m}$ ; (b)  $5x\mathbf{a}_x \text{ V/m}$ ; (c)  $5x\mathbf{a}_x + 5y\mathbf{a}_y \text{ V/m}$ .

**D4.3.** We will see later that a time-varying  $\mathbf{E}$  field need not be conservative. (If it is not conservative, the work expressed by Eq. (3) may be a function of the path used.) Let  $\mathbf{E} = y\mathbf{a}_x \text{ V/m}$  at a certain instant of time, and calculate the work required to move a 3-C charge from  $(1, 3, 5)$  to  $(2, 0, 3)$  along the straight-line segments joining: (a)  $(1, 3, 5)$  to  $(2, 3, 5)$  to  $(2, 0, 5)$  to  $(2, 0, 3)$ ; (b)  $(1, 3, 5)$  to  $(1, 3, 3)$  to  $(1, 0, 3)$  to  $(2, 0, 3)$ .

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